Chapter 1

Introduction to Statistics

§ 1.1

An Overview of Statistics

Data and Statistics

Data consists of information coming from observations, counts, measurements, or responses.

Statistics is the science of collecting, organizing, analyzing, and interpreting data in order to make decisions.

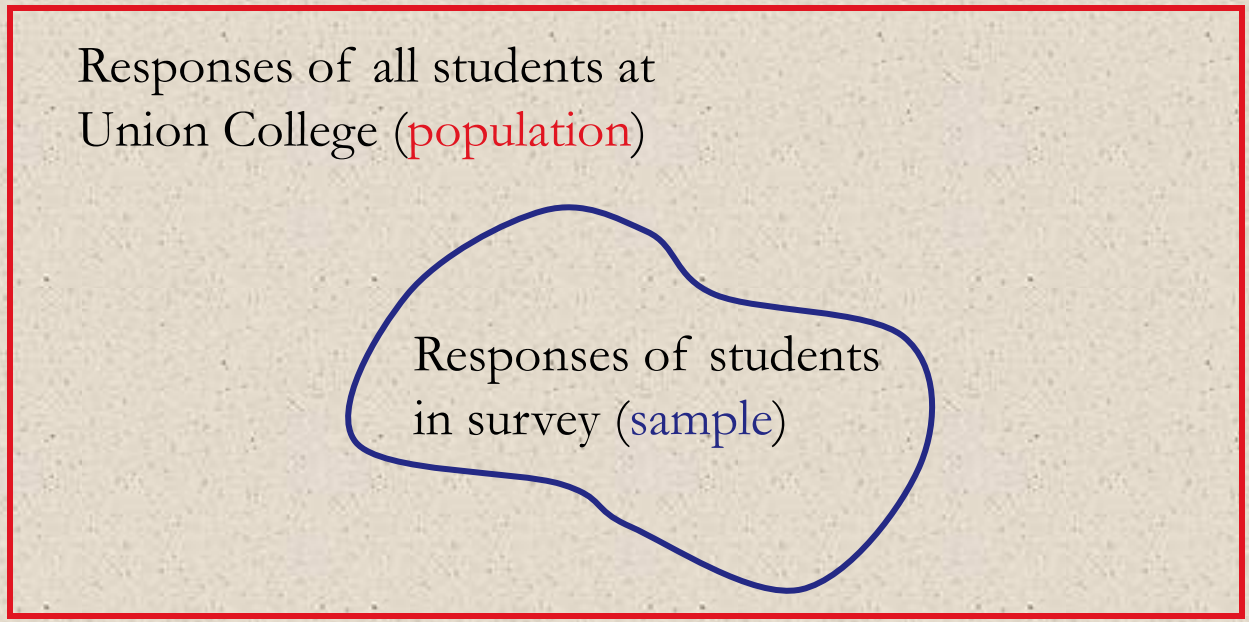
A **population** is the collection of *all* outcomes, responses, measurement, or counts that are of interest.

A **sample** is a subset of a population.

Populations & Samples

Example:

In a recent survey, 250 college students at Union College were asked if they smoked cigarettes regularly. 35 of the students said yes. Identify the population and the sample.



Parameters & Statistics

A **parameter** is a numerical description of a *population* characteristic.

A **statistic** is a numerical description of a *sample* characteristic.

Parameter → **Population**

Statistic → **Sample**

Parameters & Statistics

Example:

Decide whether the numerical value describes a population parameter or a sample statistic.

a.) A recent survey of a sample of 450 college students reported that the average weekly income for students is $325.

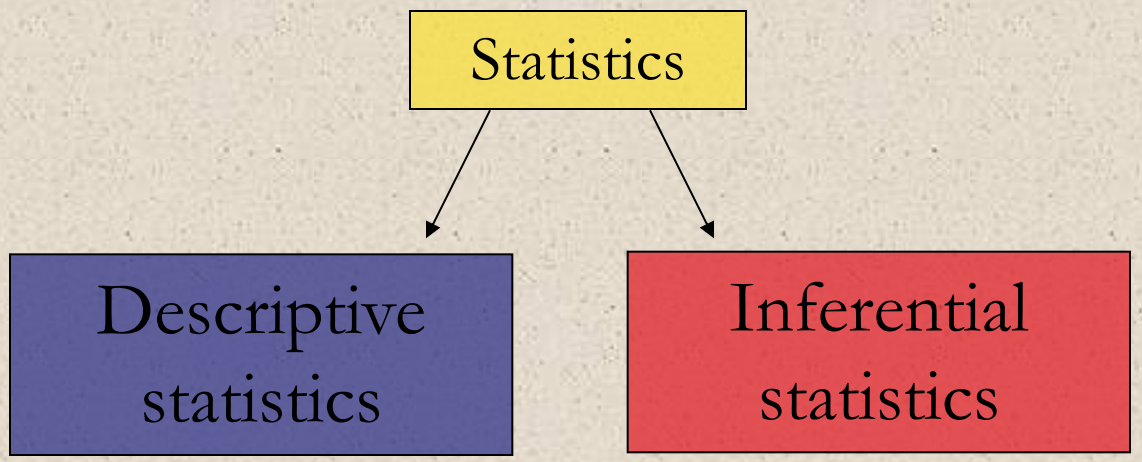
Because the average of $325 is based on a sample, this is a sample statistic.

b.) The average weekly income for all students is $405.

Because the average of $405 is based on a population, this is a population parameter.

Branches of Statistics

The study of statistics has two major branches: **descriptive statistics** and **inferential statistics**.



Descriptive and Inferential Statistics

Example:

In a recent study, volunteers who had less than 6 hours of sleep were four times more likely to answer incorrectly on a science test than were participants who had at least 8 hours of sleep. Decide which part is the descriptive statistic and what conclusion might be drawn using inferential statistics.

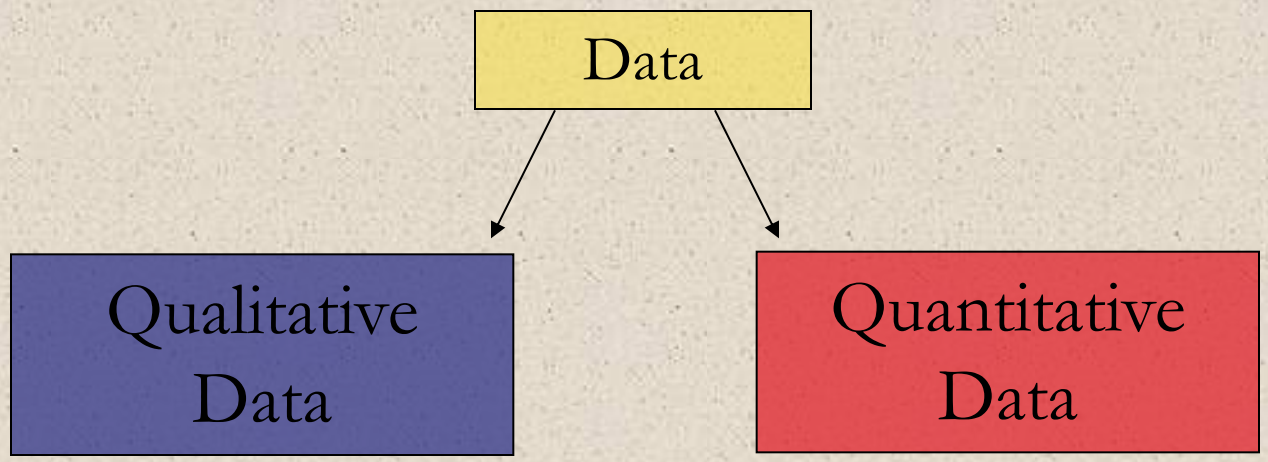
The statement "four times more likely to answer incorrectly" is a descriptive statistic. An inference drawn from the sample is that all individuals sleeping less than 6 hours are more likely to answer science question incorrectly than individuals who sleep at least 8 hours.

§ 1.2

Data Classification

Types of Data

Data sets can consist of two types of data: **qualitative data** and **quantitative data**.



Qualitative and Quantitative Data

Example:

The grade point averages of five students are listed in the table. Which data are qualitative data and which are quantitative data?

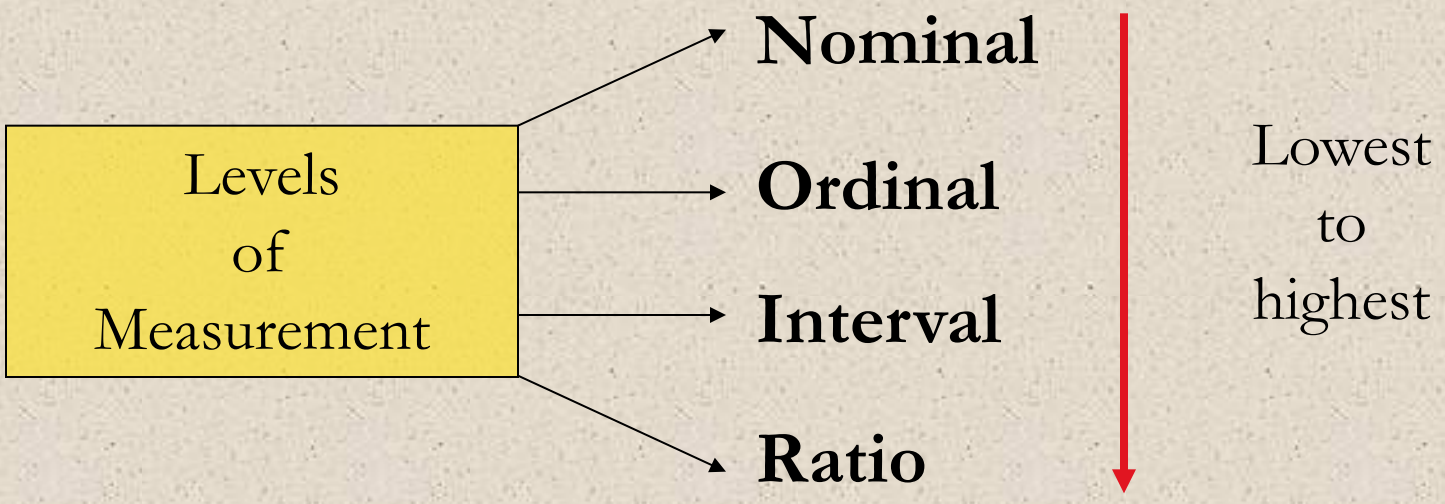
Student	GPA
Sally	3.22
Bob	3.98
Cindy	2.75
Mark	2.24
Kathy	3.84

Qualitative data

Quantitative data

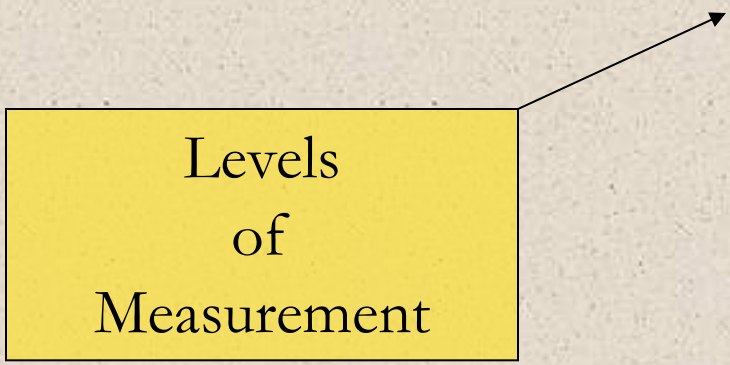
Levels of Measurement

The level of measurement determines which statistical calculations are meaningful. The four levels of measurement are: **nominal**, **ordinal**, **interval**, and **ratio**.



Nominal Level of Measurement

Data at the **nominal level of measurement** are qualitative only.



Nominal

Calculated using names, labels, or qualities. No mathematical computations can be made at this level.

Colors in the US flag

Names of students in your class

Textbooks you are using this semester

Ordinal Level of Measurement

Data at the **ordinal level of measurement** are qualitative or quantitative.

Levels of Measurement → **Ordinal**

Arranged in order, but differences between data entries are not meaningful.

Class standings: freshman, sophomore, junior, senior

Numbers on the back of each player's shirt

Top 50 songs played on the radio

Interval Level of Measurement

Data at the **interval level of measurement** are quantitative. A zero entry simply represents a position on a scale; the entry is not an inherent zero.

Levels of Measurement → **Interval**

Arranged in order, the differences between data entries can be calculated.

Temperatures

Years on a timeline

Atlanta Braves World Series victories

Ratio Level of Measurement

Data at the **ratio level of measurement** are similar to the interval level, but a zero entry is meaningful.

Levels of Measurement

A ratio of two data values can be formed so one data value can be expressed as a ratio.

Ratio

Ages

Grade point averages

Weights

Summary of Levels of Measurement

Level of measurement	Put data in categories	Arrange data in order	Subtract data values	Determine if one data value is a multiple of another
Nominal	Yes	No	No	No
Ordinal	Yes	Yes	No	No
Interval	Yes	Yes	Yes	No
Ratio	Yes	Yes	Yes	Yes

§ 1.3

Experimental Design

Designing a Statistical Study

GUIDELINES

1. Identify the variable(s) of interest (the focus) and the population of the study.
2. Develop a detailed plan for collecting data. If you use a sample, make sure the sample is representative of the population.
3. Collect the data.
4. Describe the data.
5. Interpret the data and make decisions about the population using inferential statistics.
6. Identify any possible errors.

Methods of Data Collection

In an **observational study**, a researcher observes and measures characteristics of interest of part of a population.

In an **experiment**, a treatment is applied to part of a population, and responses are observed.

A **simulation** is the use of a mathematical or physical model to reproduce the conditions of a situation or process.

A **survey** is an investigation of one or more characteristics of a population.

→ A **census** is a measurement of an *entire* population.

→ A **sampling** is a measurement of *part* of a population.

Stratified Samples

A **stratified sample** has members from each segment of a population. This ensures that each segment from the population is represented.

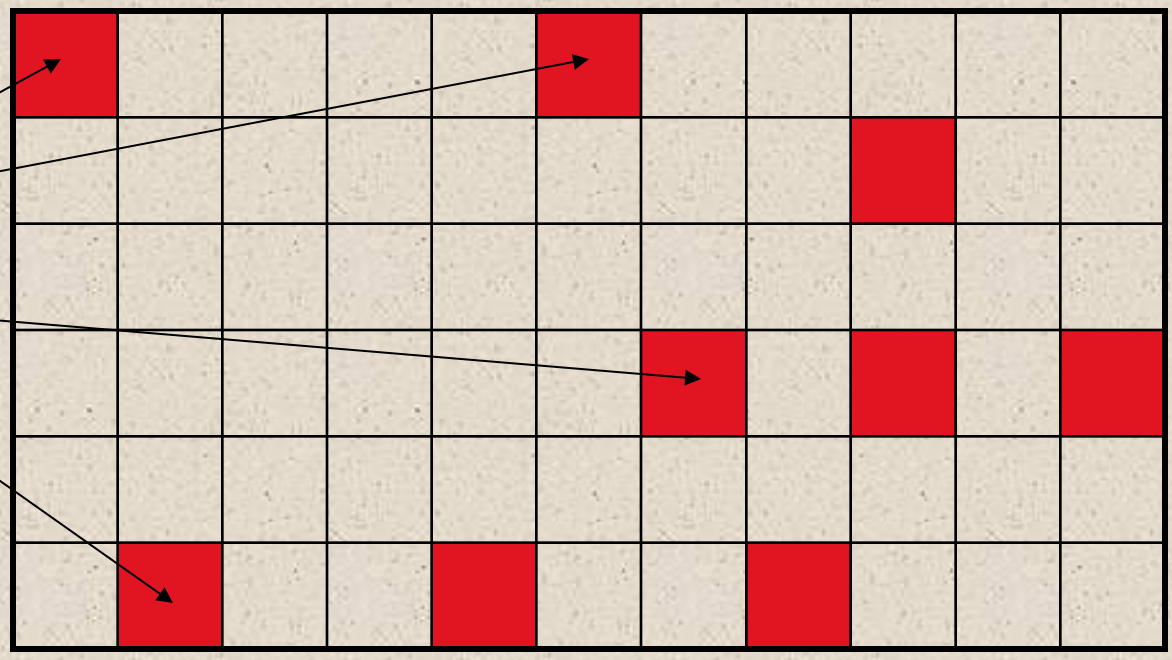


Freshmen Sophomores Juniors Seniors

Cluster Samples

A **cluster sample** has all members from randomly selected segments of a population. This is used when the population falls into naturally occurring subgroups.

All members in each selected group are used.



The city of Clarksville divided into city blocks.

Systematic Samples

A **systematic sample** is a sample in which each member of the population is assigned a number. A starting number is randomly selected and sample members are selected at regular intervals.



Every fourth member is chosen.

Convenience Samples

A **convenience sample** consists only of available members of the population.

Example:

You are doing a study to determine the number of years of education each teacher at your college has. Identify the sampling technique used if you select the samples listed.

1.) You randomly select two different departments and survey each teacher in those departments.

2.) You select only the teachers you currently have this semester.

3.) You divide the teachers up according to their department and then choose and survey some teachers in each department.

Continued.

Identifying the Sampling Technique

Example continued:

You are doing a study to determine the number of years of education each teacher at your college has. Identify the sampling technique used if you select the samples listed.

1.) This is a cluster sample because each department is a naturally occurring subdivision.

2.) This is a convenience sample because you are using the teachers that are readily available to you.

3.) This is a stratified sample because the teachers are divided by department and some from each department are randomly selected.

§ 1.4

Descriptive Statistics

Descriptive Statistics

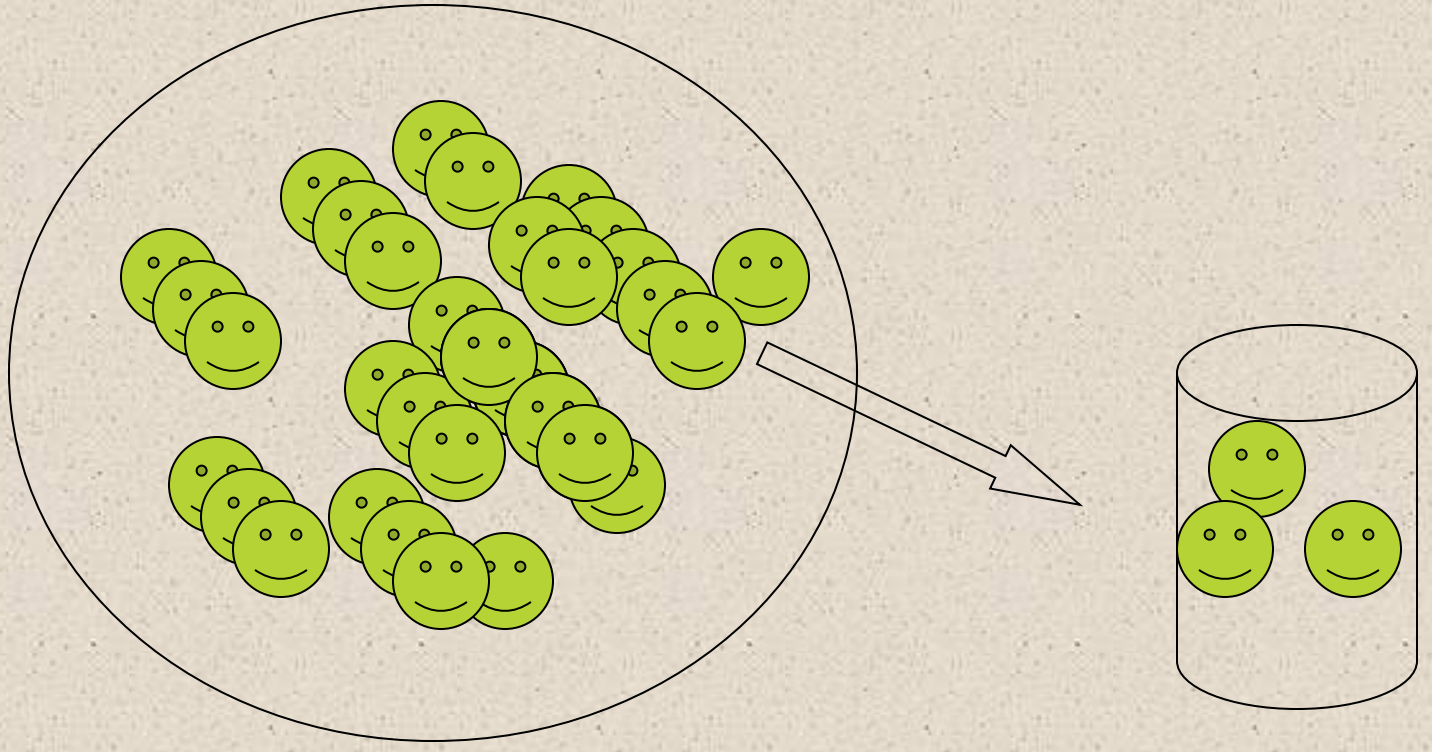
Descriptive Statistics are Used by Researchers to Report on Populations and Samples

In Sociology:

Summary descriptions of measurements (variables) taken about a group of people

By Summarizing Information, Descriptive Statistics Speed Up and Simplify Comprehension of a Group's Characteristics

Sample vs. Population



Descriptive Statistics

An Illustration:

Which Group is Smarter?

Class A--IQs of 13 Students	
102	115
128	109
131	89
98	106
140	119
93	97
110	

Class B--IQs of 13 Students	
127	162
131	103
96	111
80	109
93	87
120	105
109	

Each individual may be different. If you try to understand a group by remembering the qualities of each member, you become overwhelmed and fail to understand the group.

Descriptive **Statistics**

Which group is smarter now?

Class A--Average IQ	**Class B--Average IQ**
110.54	110.23

They're roughly the same!

With a summary descriptive statistic, it is much easier to answer our question.

Descriptive Statistics

Types of descriptive statistics:

Organize Data

- Tables
- Graphs

Summarize Data

- Central Tendency
- Variation

SPSS Output for Frequency Distribution

IQ

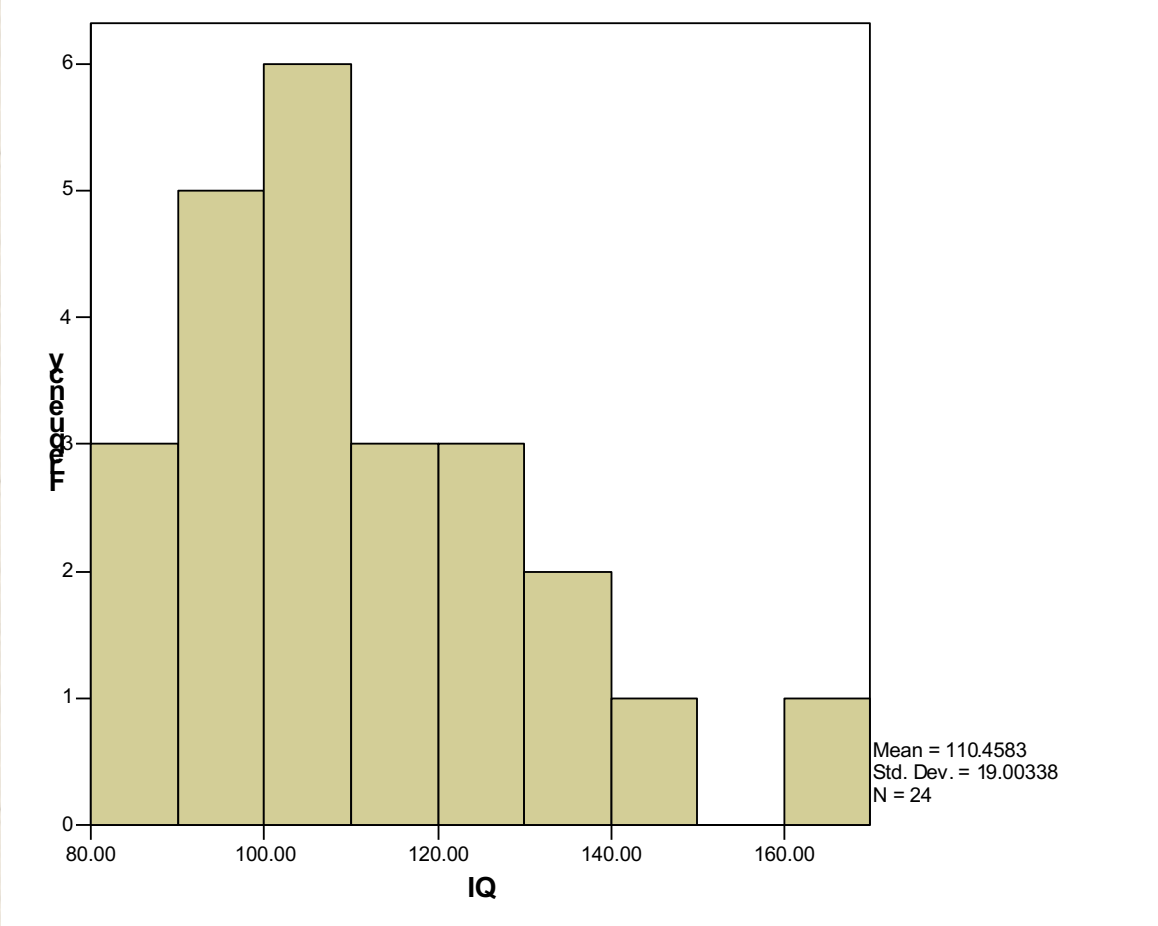
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	82.00	1	4.2	4.2	4.2
	87.00	1	4.2	4.2	8.3
	89.00	1	4.2	4.2	12.5
	93.00	2	8.3	8.3	20.8
	96.00	1	4.2	4.2	25.0
	97.00	1	4.2	4.2	29.2
	98.00	1	4.2	4.2	33.3
	102.00	1	4.2	4.2	37.5
	103.00	1	4.2	4.2	41.7
	105.00	1	4.2	4.2	45.8
	106.00	1	4.2	4.2	50.0
	107.00	1	4.2	4.2	54.2
	109.00	1	4.2	4.2	58.3
	111.00	1	4.2	4.2	62.5
	115.00	1	4.2	4.2	66.7
	119.00	1	4.2	4.2	70.8
	120.00	1	4.2	4.2	75.0
	127.00	1	4.2	4.2	79.2
	128.00	1	4.2	4.2	83.3
	131.00	2	8.3	8.3	91.7
	140.00	1	4.2	4.2	95.8
	162.00	1	4.2	4.2	100.0
	Total	24	100.0	100.0	

Relative Frequency Distribution

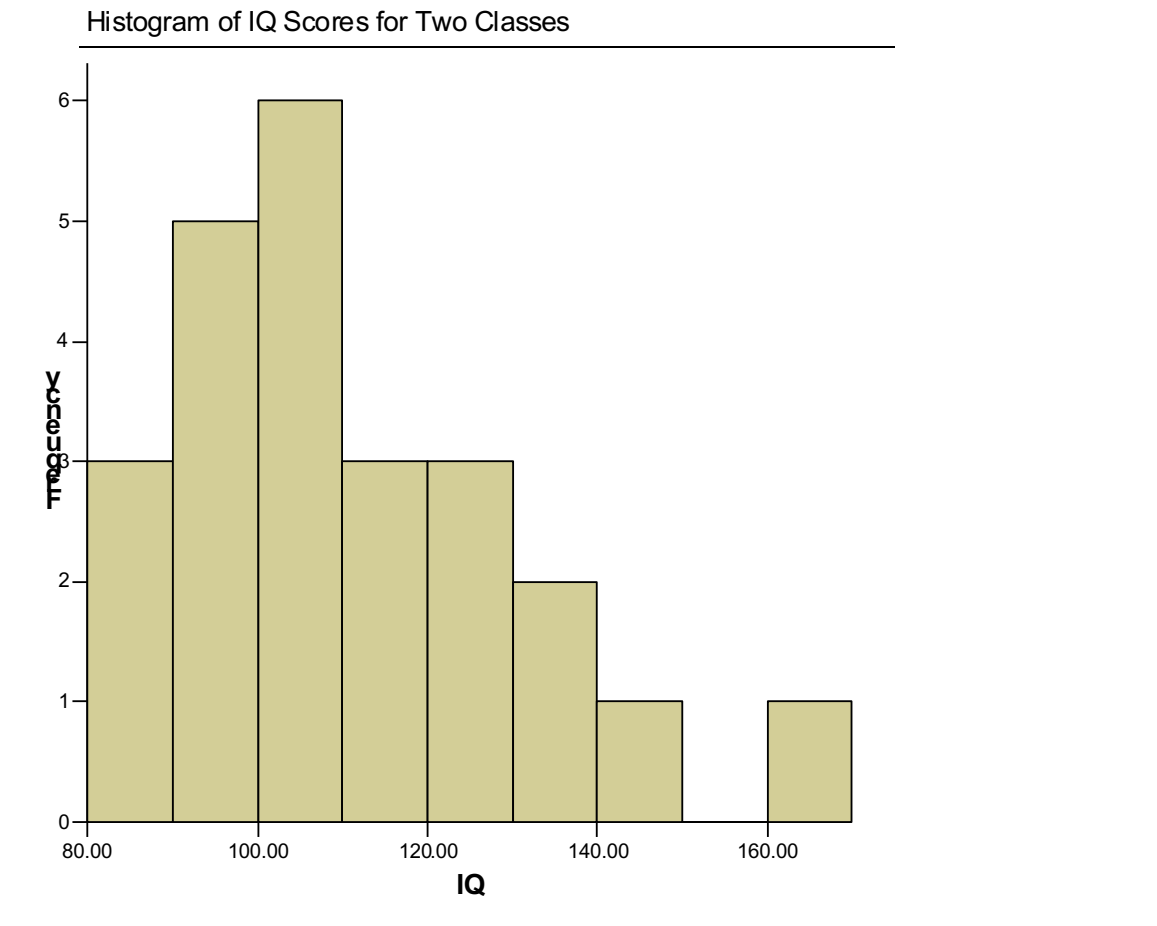
Relative Frequency Distribution of IQ for Two Classes

IQ	Frequency	Percent	Valid Percent	Cumulative Percent
82.00	1	4.2	4.2	4.2
87.00	1	4.2	4.2	8.3
89.00	1	4.2	4.2	12.5
93.00	2	8.3	8.3	20.8
96.00	1	4.2	4.2	25.0
97.00	1	4.2	4.2	29.2
98.00	1	4.2	4.2	33.3
102.00	1	4.2	4.2	37.5
103.00	1	4.2	4.2	41.7
105.00	1	4.2	4.2	45.8
106.00	1	4.2	4.2	50.0
107.00	1	4.2	4.2	54.2
109.00	1	4.2	4.2	58.3
111.00	1	4.2	4.2	62.5
115.00	1	4.2	4.2	66.7
119.00	1	4.2	4.2	70.8
120.00	1	4.2	4.2	75.0
127.00	1	4.2	4.2	79.2
128.00	1	4.2	4.2	83.3
131.00	2	8.3	8.3	91.7
140.00	1	4.2	4.2	95.8
162.00	1	4.2	4.2	100.0
Total	24	100.0	100.0	

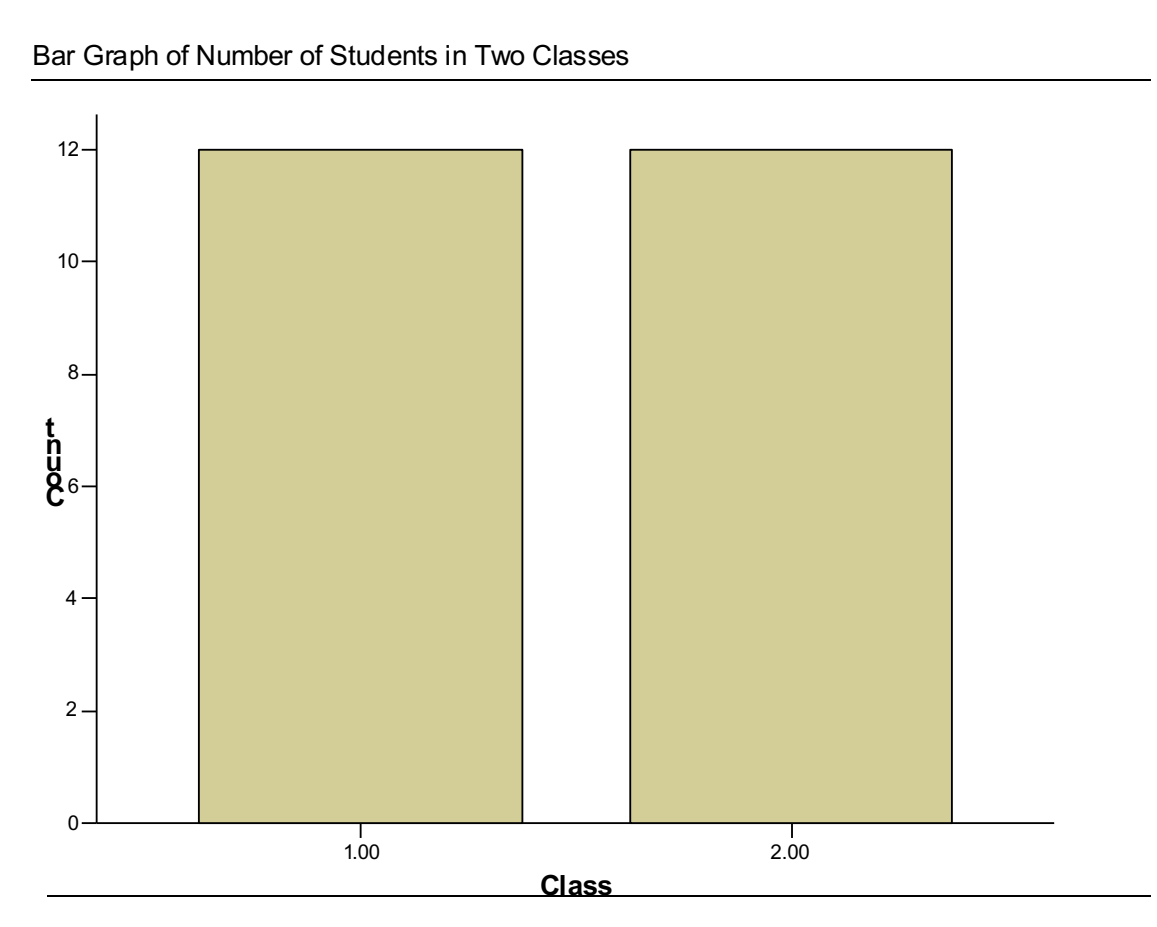
SPSS Output for Histogram



Histogram


Histogram of IQ Scores for Two Classes
6
5
4
3
2
1
0
80.00
100.00
120.00
140.00
160.00
IQ

Bar Graph


Bar Graph of Number of Students in Two Classes
12
10
8
6
4
2
0
Count
1.00
2.00
Class

Mean

Most commonly called the “average.”

Add up the values for each case and divide by the total number of cases.

$$\text{Y-bar} = \frac{(Y1 + Y2 + \ldots + Yn)}{n}$$

$$\text{Y-bar} = \frac{\Sigma Y_i}{n}$$

Mean



What's up with all those symbols, man?

$$\text{Y-bar} = \frac{(Y1 + Y2 + \ldots + Yn)}{n}$$

$$\text{Y-bar} = \frac{\Sigma\, Y_i}{n}$$

Some Symbolic Conventions in this Class:

Y = your variable (could be X or Q or ☺ or even "Glitter")

"-bar" or line over symbol of your variable = mean of that variable

Y1 = first case's value on variable Y

". . ." = ellipsis = continue sequentially

Yn = last case's value on variable Y

n = number of cases in your sample

Σ = Greek letter "sigma" = sum or add up what follows

i = a typical case or each case in the sample (1 through n)

Mean

Class A--IQs of 13 Students

102	115		
128	109		
131	89		
98		106	
140	119		
93		97	
110			

$\Sigma Y_i = 1437$

$\text{Y-bar}_A = \frac{\Sigma Y_i}{n} = \frac{1437}{13} = 110.54$

Class B--IQs of 13 Students

127	162		
131	103		
96		111	
80		109	
93		87	
120	105		
109			

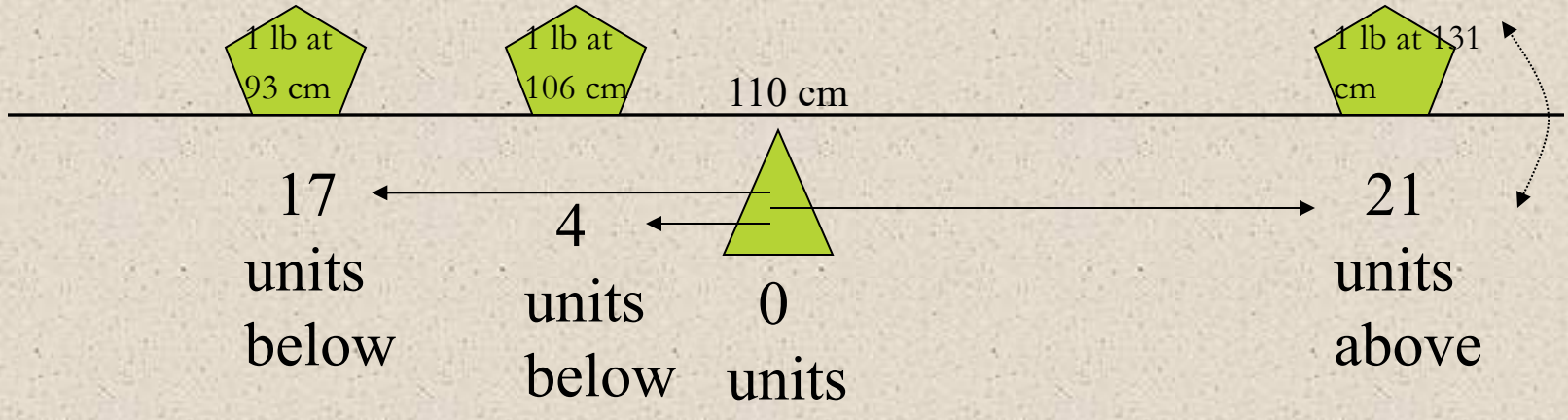
$\Sigma Y_i = 1433$

$\text{Y-bar}_B = \frac{\Sigma Y_i}{n} = \frac{1433}{13} = 110.23$

Mean

The mean is the "balance point."

Each person's score is like 1 pound placed at the score's position on a see-saw. Below, on a 200 cm see-saw, the mean equals 110, the place on the see-saw where a fulcrum finds balance:

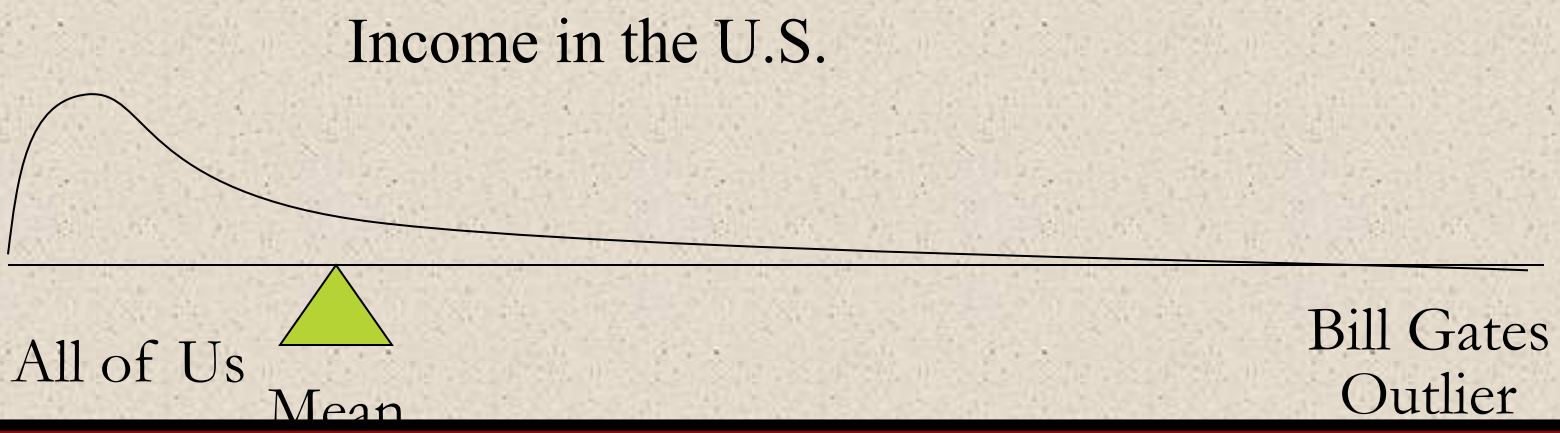


The scale is balanced because…

$17 + 4 = 21$

Mean

1. Means can be badly affected by outliers (data points with extreme values unlike the rest)
2. Outliers can make the mean a bad measure of central tendency or common experience



Median

The middle value when a variable's values are ranked in order; the point that divides a distribution into two equal halves.

When data are listed in order, the median is the point at which 50% of the cases are above and 50% below it.

The 50th percentile.

Median

Class A--IQs of 13 Students

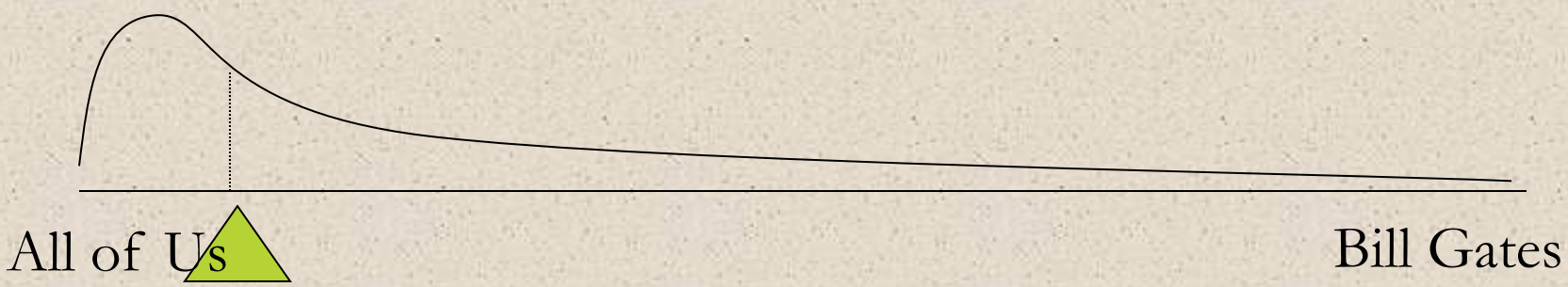
89
93
97
98
102
106
109
110
115
119
128
131
140

Median = 109

(six cases above, six below)

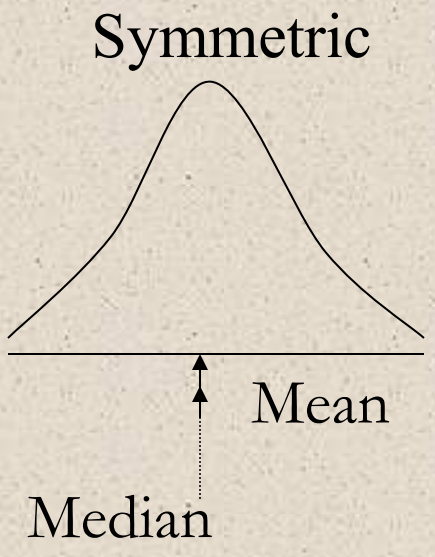
Median

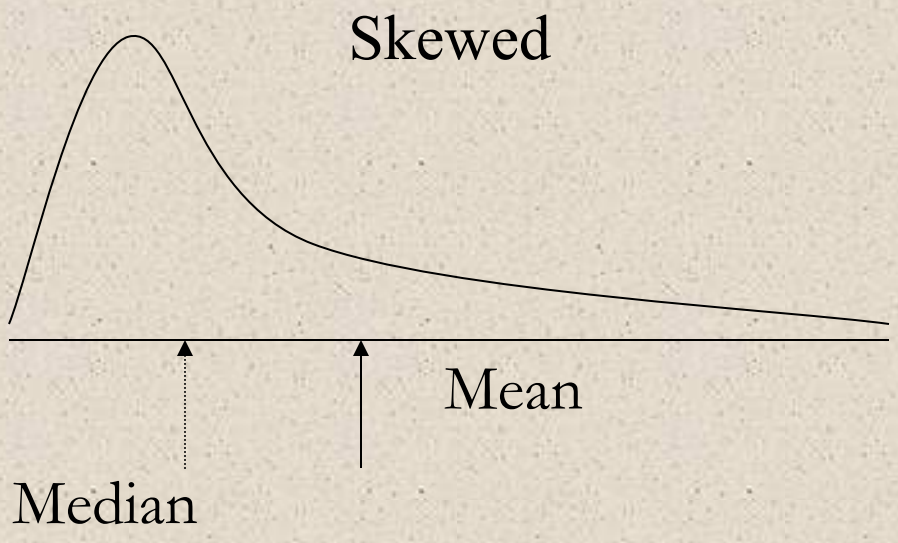
1. The median is unaffected by outliers, making it a better measure of central tendency, better describing the "typical person" than the mean when data are skewed.



Median

2. If the recorded values for a variable form a symmetric distribution, the median and mean are identical.
3. In skewed data, the mean lies further toward the skew than the median.





Median

The middle score or measurement in a set of ranked scores or measurements; the point that divides a distribution into two equal halves.

Data are listed in order—the median is the point at which 50% of the cases are above and 50% below.

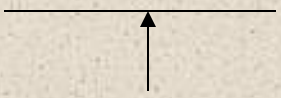
The 50th percentile.

Mode

The most common data point is called the mode.

The combined IQ scores for Classes A & B:

80 87 89 93 93 96 97 98 102 103 105 106 109 109 109 110 111 115 119 120
127 128 131 131 140 162

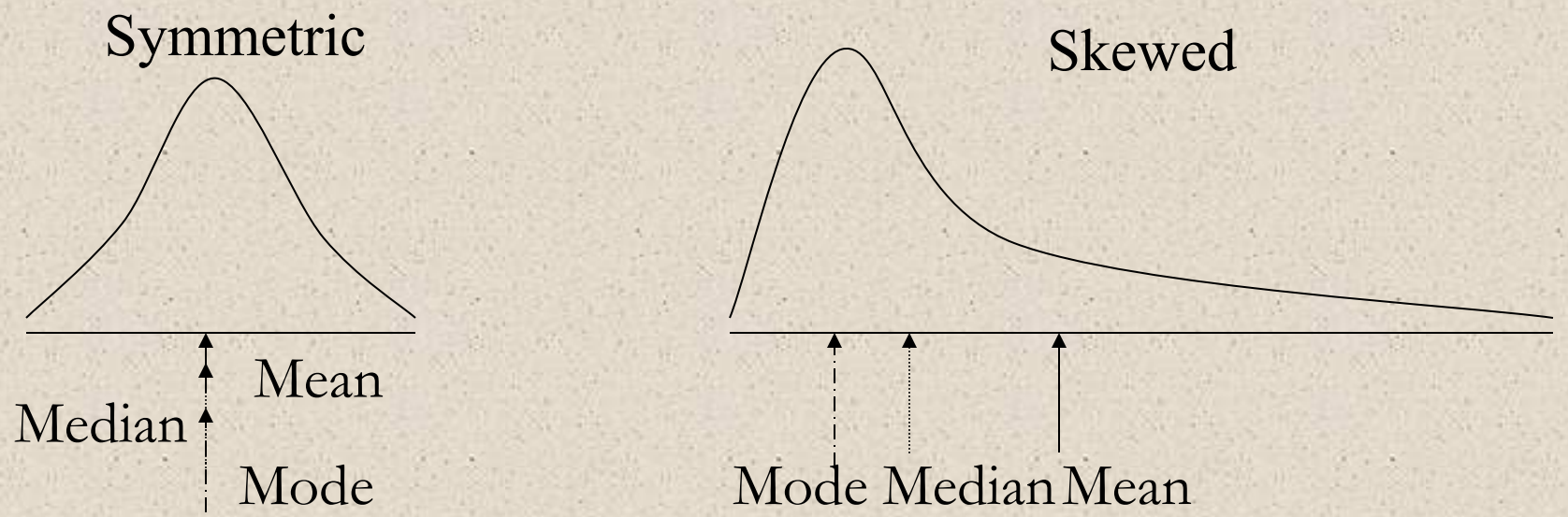


A la mode!!

BTW, It is possible to have more than one mode!

Mode

1. It may give you the most likely experience rather than the "typical" or "central" experience.
2. In symmetric distributions, the mean, median, and mode are the same.
3. In skewed data, the mean and median lie further toward the skew than the mode.



Range

The spread, or the distance, between the lowest and highest values of a variable.

To get the range for a variable, you subtract its lowest value from its highest value.

Class A--IQs of 13 Students

102	115
128	109
131	89
98	106
140	119
93	97
110	

Class A Range = 140 - 89 = 51

Class B--IQs of 13 Students

127	162
131	103
96	111
80	109
93	87
120	105
109	

Class B Range = 162 - 80 = 82

Interquartile Range

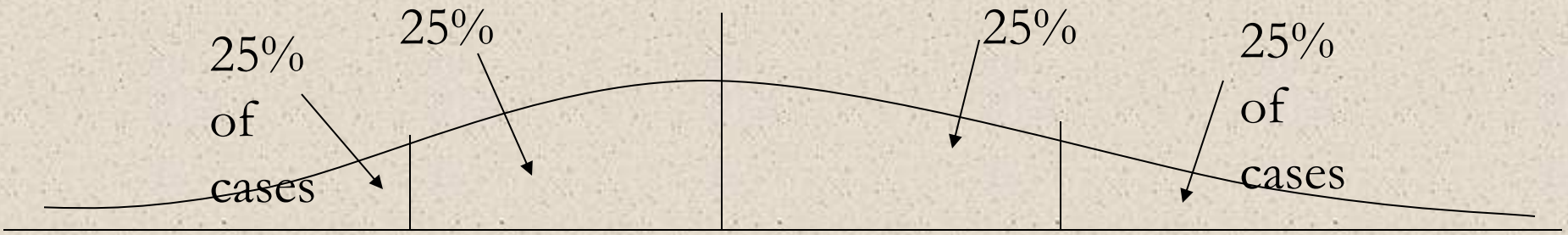
A quartile is the value that marks one of the divisions that breaks a series of values into four equal parts.

The median is a quartile and divides the cases in half.

25th percentile is a quartile that divides the first ¼ of cases from the latter ¾.

75th percentile is a quartile that divides the first ¾ of cases from the latter ¼.

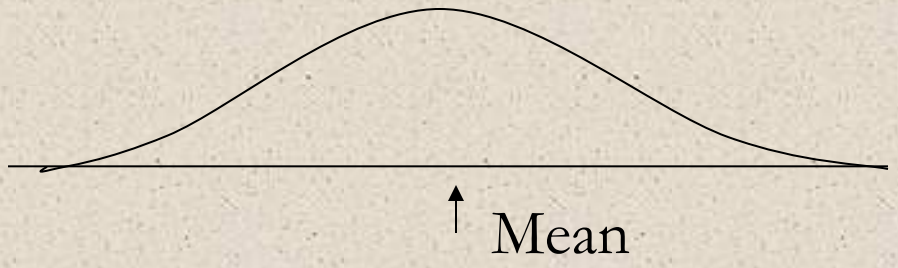
The interquartile range is the distance or range between the 25th percentile and the 75th percentile. Below, what is the interquartile range?



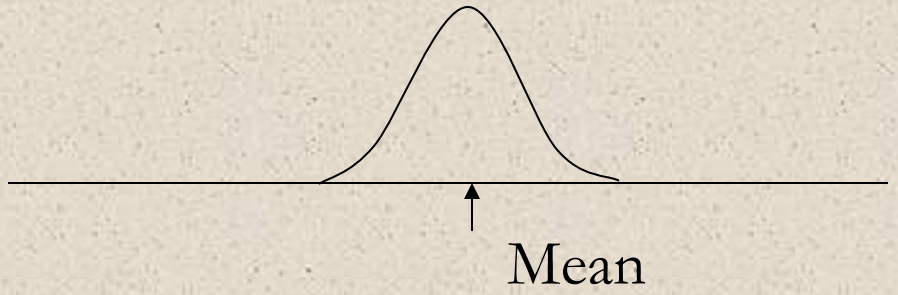
Variance

A measure of the spread of the recorded values on a variable. A measure of dispersion.

The larger the variance, the further the individual cases are from the mean.



The smaller the variance, the closer the individual scores are to the mean.



Variance

Variance is a number that at first seems complex to calculate.

Calculating variance starts with a "deviation."

A deviation is the distance away from the mean of a case's score.

Y_i – Y-bar

If the average person's car costs $20,000, my deviation from the mean is - $14,000!

6K - 20K = -14K

Variance

The deviation of 102 from 110.54 is? Deviation of 115?

Class A--IQs of 13 Students	
102	115
128	109
131	89
98	106
140	119
93	97
110	

$\text{Y-bar}_A = 110.54$

Variance

The deviation of 102 from 110.54 is? Deviation of 115?

102 - 110.54 = -8.54 115 - 110.54 = 4.46

Class A--IQs of 13 Students

102	115
128	109
131	89
98	106
140	119
93	97
110	

Y-bar_A ***= 110.54***

Variance

We want to add these to get total deviations, but if we were to do that, we would get zero every time. Why?

We need a way to eliminate negative signs.

Squaring the deviations will eliminate negative signs...

A Deviation Squared: $(Yi - \text{Y-bar})^2$

Back to the IQ example,

A deviation squared for 102 is: $(102 - 110.54)^2 = (-8.54)^2 = 72.93$

of 115: $(115 - 110.54)^2 = (4.46)^2 = 19.89$

Variance

If you were to add all the squared deviations together, you'd get what we call the

"Sum of Squares."

Sum of Squares (SS) = $\Sigma (Yi - \text{Y-bar})^2$

$SS = (Y1 - \text{Y-bar})^2 + (Y2 - \text{Y-bar})^2 + \ldots + (Yn - \text{Y-bar})^2$

Variance

Class A, sum of squares:

$(102 - 110.54)^2 + (115 - 110.54)^2 +$
$(126 - 110.54)^2 + (109 - 110.54)^2 +$
$(131 - 110.54)^2 + (89 - 110.54)^2 +$
$(98 - 110.54)^2 + (106 - 110.54)^2 +$
$(140 - 110.54)^2 + (119 - 110.54)^2 +$
$(93 - 110.54)^2 + (97 - 110.54)^2 +$
$(110 - 110.54) = SS = 2825.39$

Class A--IQs of 13 Students

102	115
128	109
131	89
98	106
140	119
93	97
110	

Y-bar = 110.54

Variance

The last step…

The approximate average sum of squares is the variance.

SS/N = Variance for a population.

SS/n-1 = Variance for a sample.

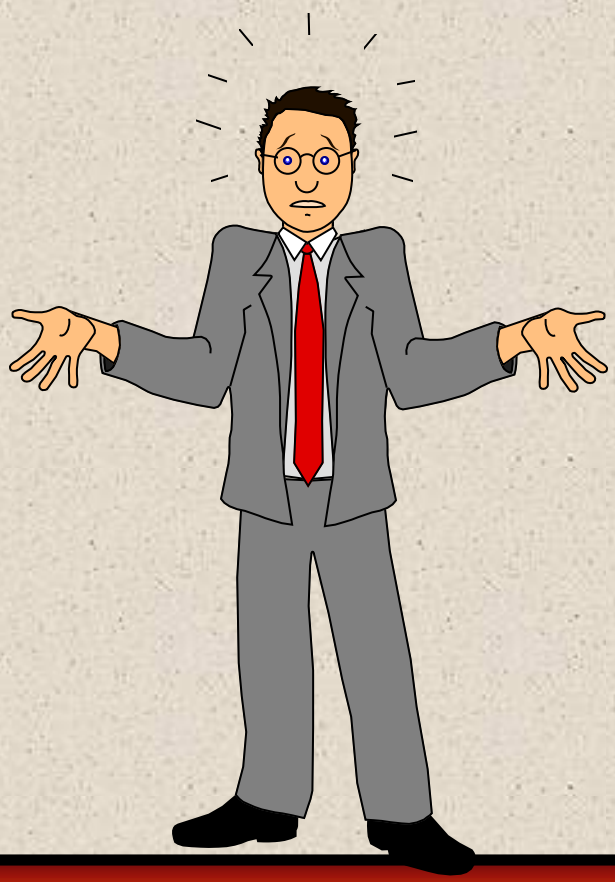
Variance = $\Sigma(Yi - \text{Y-bar})^2 / n - 1$

Variance

For Class A, Variance = 2825.39 / n - 1

= 2825.39 / 12 = 235.45

How helpful is that???



Standard Deviation

To convert variance into something of meaning, let's create standard deviation.

The square root of the variance reveals the average deviation of the observations from the mean.

$$\text{s.d.} = \sqrt{\frac{\Sigma(Yi - \text{Y-bar})^2}{n - 1}}$$

Standard Deviation

For Class A, the standard deviation is:

$$\sqrt{235.45} = 15.34$$

The average of persons' deviation from the mean IQ of 110.54 is 15.34 IQ points.

Review:

1. Deviation
2. Deviation squared
3. Sum of squares
4. Variance
5. Standard deviation

Standard Deviation

1. Larger s.d. = greater amounts of variation around the mean.

   For example:



$\overline{Y} = 25$

s.d. = 3

$\overline{Y} = 25$

s.d. = 6

2. s.d. = 0 only when all values are the same (only when you have a constant and not a "variable")
3. If you were to "rescale" a variable, the s.d. would change by the same magnitude—if we changed units above so the mean equaled 250, the s.d. on the left would be 30, and on the right, 60
4. Like the mean, the s.d. will be inflated by an outlier case value.

Standard Deviation

Note about computational formulas:

Your book provides a useful short-cut formula for computing the variance and standard deviation.

This is intended to make hand calculations as quick as possible.

They obscure the conceptual understanding of our statistics. SPSS and the computer are "computational formulas" now.

Box-Plots

A way to graphically portray almost all the descriptive statistics at once is the box-plot.

A box-plot shows: Upper and lower quartiles

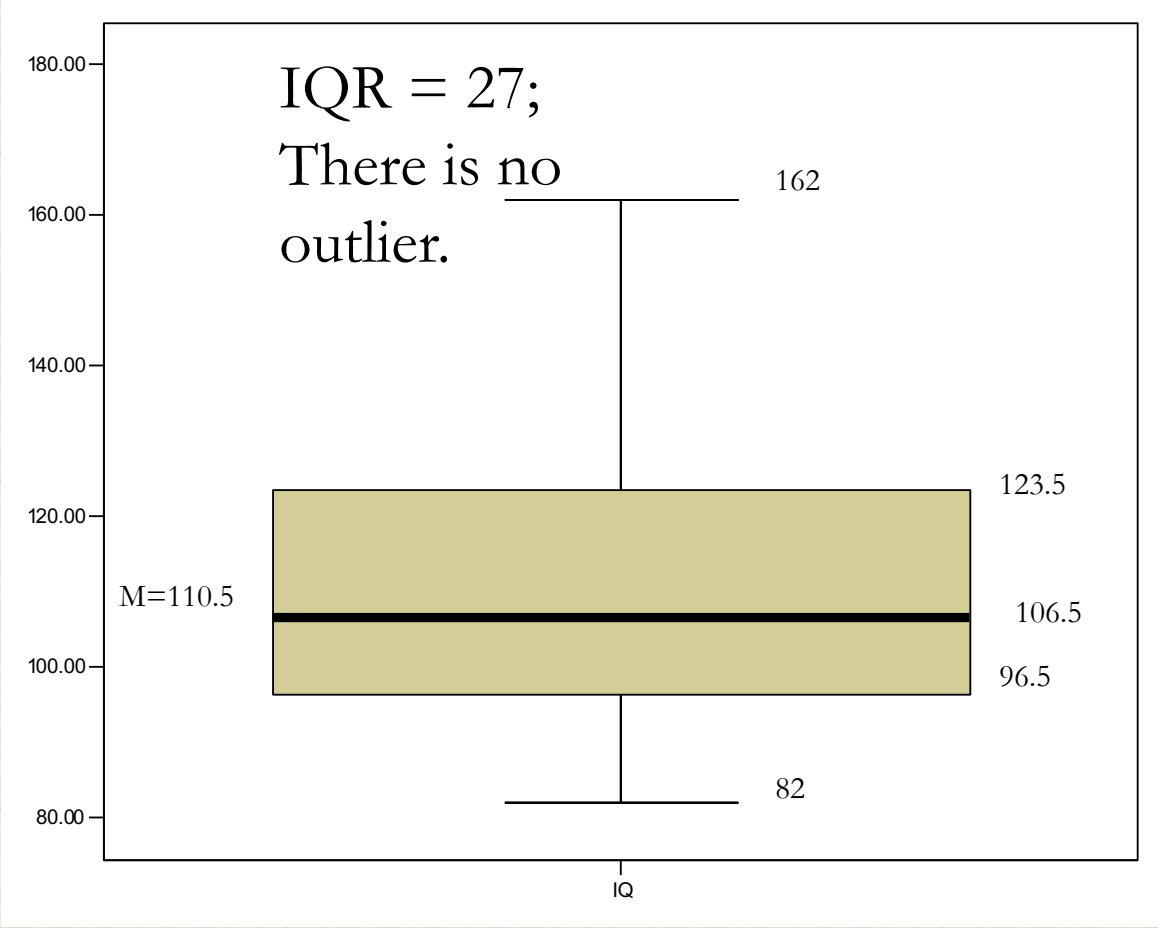
Mean

Median

Range

Outliers (1.5 IQR)

Box-Plots


IQR = 27;
There is no outlier.
180.00
160.00
140.00
120.00
100.00
80.00
162
123.5
M=110.5
106.5
96.5
82
IQ

Descriptive Statistics

Now you are qualified use descriptive statistics!

Questions?

